

Agent Evaluation with MLFlow and EvalHub on Red Hat OpenShift AI

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Home

Goal

This course teaches how to deploy MLflow and EvalHub as included with Red Hat OpenShift AI 3.4; how to run evaluations of a self-hosted model, using EvalHub; how to run evaluations of an agentic application, using the MLflow SDK with Python; and how to visualize the results of those evaluations as MLflow experiments and traces using the Red Hat OpenShift AI dashboard.

Total estimated reading time: ...

You should reserve between 4 hours to one working day to complete this course.



Red Hat associates must enroll using the [Red Hat Learning Portal](#), and Red Hat Partners must enroll using [Red Hat Partner Connect](#) to register this course in their learning history.

Objectives

On completing this course, you should be able to:

- Understand how MLflow and EvalHub relate to agentic applications and to Red Hat OpenShift AI
- Deploy a development environment with MLflow and EvalHub
- Evaluate LLMs using EvalHub
- Evaluate agents using MLflow

Audience

- Software developers working on agentic applications and agentic harnesses.
- Red Hat and Partner roles that perform demonstrations, PoC and PoV of MLflow with Red Hat OpenShift AI, such as Solutions Architects.
- Red Hat and Partner roles that implement and support MLflow and EvalHub with Red Hat OpenShift AI, such as Services Consultants, Technical Account Managers, and Customer Support Engineers.

Prerequisites

This course assumes that you have the following experience:

- Familiarity with generative AI and agentic application concepts.
- Familiarity with the basic operation of Red Hat OpenShift AI.
- Python programming skills are recommended but not required, because this training provides sample code which is ready to run.

It is recommended, but not required, to attend a prerequisite course about agentic traces with MLflow.

Classroom environment

This course uses the [Red Hat OpenShift AI 3.4](#) demo catalog entry, which provides an SNO OpenShift cluster based on an AWS instance with a single GPU and a predeployed llama-32-3b-instruct model.

Any additional components required by MLflow, EvalHub, or course activities are configured by learners as part of course activities, using sample Python code and Kubernetes manifests provided in a public Git repository.

Other sources of information about MLflow and EvalHub

For documentation about Red Hat OpenShift AI, see its [product documentation](#), especially:

- [Working with MLflow](#)
- Chapter 2 of Evaluating AI systems: [Evaluate LLMs with EvalHub](#)

Also refer to upstream documentation of <https://mlflow.org/docs/latest/index.html> [MLflow^] and [EvalHub](#).

Author

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Lab Environment

FIXME: Instructions for accessing the lab environment for this hands-on based training.

Introduction to MLFlow and EvalHub

Goal

Understand how MLFlow and EvalHub relate to agentic applications and to Red Hat OpenShift AI.



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Evaluation Concepts for AI Applications

Objective

Describe how evaluating AI-enabled applications differs from testing traditional applications and identify the roles of MLFlow and EvalHub in Red Hat OpenShift AI.



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What's Next

Now that you understand the key evaluation concepts for AI applications, test your knowledge with a quiz on MLFlow and EvalHub fundamentals.

Quiz: MLFlow and EvalHub Concepts Review

Objective

Verify your understanding of key MLFlow and EvalHub concepts introduced in the previous section.

Interactive Assessment

What's next

Next, you explore the Red Hat OpenShift AI demo environment and verify that the predeployed model is ready for evaluation activities.

Lab: Explore the Learning Environment

Objective

Navigate the provided demo environment, identify enabled Red Hat OpenShift AI features, and verify that the predeployed model is functional.



Not started yet.

Before you begin

Describe requirements, such as customer portal accounts, Red Hat Developer free sub, already configured machines, access to container registries, etc.

Instructions

High-level summary of what you will perform in this lab, but more detailed than the objective.

1. Task or outcome.
 - a. Something to do, using a graphical or web UI. Click **Save As** to save your work.
 - b. More things to do
2. Task or outcome.
 - a. Something to do, using a shell.

```
$ command --options fixed-arguments variable-arguments
output of the command
highlight something in the output
```

- b. More things to do. Press `Ctrl` + `C` to interrupt.

Summary of what you accomplished.

What's next

Now that you are familiar with the learning environment, the next chapter covers how to deploy MLFlow and EvalHub for development on Red Hat OpenShift AI.

Deploy MLFlow and EvalHub

Goal

Deploy a development environment with MLFlow and EvalHub on Red Hat OpenShift AI.



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MLFlow Architecture and Integration

Objective

Describe the architecture of MLFlow and explain how to enable and access it in Red Hat OpenShift AI.



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What's Next

Next, you deploy and configure MLFlow on your Red Hat OpenShift AI instance.

Lab: Deploy MLFlow for Development

Objective

Enable MLFlow on Red Hat OpenShift AI, configure it with local storage, and verify the deployment is functional.



Not started yet.

Before you begin

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Instructions

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output of the command
highlight something in the output
```

- b. More things to do. Press `Ctrl` + `C` to interrupt.

Summary of what you accomplished.

What's next

With MLFlow deployed and verified, the next section introduces the architecture and setup of EvalHub.

EvalHub Architecture and CLI

Objective

Describe the architecture of EvalHub and explain how to enable, access, and use its CLI with Red Hat OpenShift AI.



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What's Next

Next, you deploy and configure EvalHub on your Red Hat OpenShift AI instance.

Lab: Deploy EvalHub for Development

Objective

Enable EvalHub on Red Hat OpenShift AI using the TrustyAI operator, configure its database, and install and configure the EvalHub CLI.



Not started yet.

Before you begin

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Summary of what you accomplished.

What's next

With both MLFlow and EvalHub deployed, the next chapter introduces model evaluation using EvalHub providers and benchmarks.

Model Evaluation with EvalHub

Goal

Evaluate LLMs using EvalHub providers, benchmarks, and collections on Red Hat OpenShift AI.



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Model Evaluation Concepts

Objective

Describe the evaluation providers and benchmarks available in EvalHub and explain how to manage evaluation jobs and interpret their results.



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What's Next

Next, you run evaluation jobs on a self-hosted model and inspect the results.

Lab: Evaluate a Self-Hosted Model

Objective

Run evaluation jobs on a self-hosted model using EvalHub, monitor their progress, and inspect results and metrics from the RHOAI dashboard.



Not started yet.

Before you begin

Describe requirements, such as customer portal accounts, Red Hat Developer free sub, already configured machines, access to container registries, etc.

Instructions

High-level summary of what you will perform in this lab, but more detailed than the objective.

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Summary of what you accomplished.

What's next

Having evaluated models with EvalHub, the next chapter focuses on evaluating agentic applications using the MLFlow SDK.

Agent Evaluation with MLFlow

Goal

Evaluate agentic applications using the MLFlow SDK with deterministic scorers and LLM judges on Red Hat OpenShift AI.



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Agentic Evaluation Concepts

Objective

Describe the different kinds of assessment for agentic applications and explain how to code MLFlow evaluations using scorers, judges, and data sets.



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What's Next

Next, you evaluate a chat application using a deterministic scorer and view the results as MLFlow traces.

Lab: Evaluate an Agent with Deterministic Scoring

Objective

Add tracing to a chat application, create a deterministic scorer, and run an MLFlow evaluation to verify response quality.



Not started yet.

Before you begin

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Instructions

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Summary of what you accomplished.

What's next

Next, you evaluate the same application using built-in and custom LLM judges for more nuanced assessment.

Lab: Evaluate an Agent with an LLM Judge

Objective

Perform evaluations using built-in and custom LLM judges with self-hosted models, and inspect evaluation results through MLFlow traces.



Not started yet.

Before you begin

Describe requirements, such as customer portal accounts, Red Hat Developer free sub, already configured machines, access to container registries, etc.

Instructions

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highlight something in the output

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Summary of what you accomplished.

What's next

Congratulations on completing this course! You are now able to deploy MLFlow and EvalHub on Red Hat OpenShift AI, evaluate both models and agentic applications, and visualize results using the RHOAI dashboard.